SHANTVEER KESTI

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SUMMARY

AI/ML Engineer with hands-on experience in PyTorch, TensorFlow, Scikit-learn, Pandas, and Generative AI frameworks including LangChain, LangGraph, and MCP. Built multiple end-to-end AI systems including RAG pipelines, LLM agents, and MCP-based tool servers using FastAPI, Docker, and AWS deployment.

EDUCATION

Alva's Institute of Engineering And Technology 2022-2026

- Bachelor of Engineering in Artificial Intelligence and Machine Learning (PURSUING)
- CGPA: 8.88

PROJECTS

- AI Research Agent (RAG + Long-Term Memory + Tool-Augmented LLM) - Link
  - Developed a tool-enabled AI agent using LangGraph and LangChain capable of retrieving information from documents, web sources, and research papers using Retrieval-Augmented Generation (RAG).
  - Implemented persistent conversational memory and streaming responses using FastAPI, PostgreSQL (PGVector), and LangSmith for observability, with a Streamlit-based interactive interface.
- AI Expense Tracker using Model Context Protocol (MCP) - Link
  - Developed a Local MCP Server using Python and FastMCP to enable AI assistants to manage expenses through tools such as add, list, summarize, edit, and delete, with SQLite-based storage and structured categories.
  - Designed and deployed a Remote MCP Server accessible via HTTP and integrated it with a local proxy MCP architecture, allowing AI clients to securely access remote expense management tools.
- RAG-Based AI Document Assistant using LangGraph - Link
  - Built an end-to-end Retrieval-Augmented Generation (RAG) system using LangChain, LangGraph, FastAPI, and pgvector to enable intelligent document question answering.
  - Implemented Dockerized microservices with PostgreSQL vector search and Streamlit UI, enabling scalable semantic search and contextual AI responses.
- Breast Cancer Detection Using Machine Learning - Link
  - Developed a machine learning model using Support Vector Machine (SVM) to classify breast tumors as benign or malignant based on 30 diagnostic features from the Breast Cancer Wisconsin dataset.
  - Built an interactive Streamlit web dashboard to collect medical inputs and provide real-time predictions, integrating a Scikit-learn pipeline with a serialized Pickle model.
- Data-Driven Study on the Dermatological Impact of Sunscreen Formulations - Link
  - Conducted a comparative analysis of mineral and chemical sunscreen products to assess effectiveness and dermatological safety.
  - Identified high-risk ingredients associated with skin irritation, providing actionable insights to support informed consumer decisions.

CERTIFICATIONS

- Data Structures and Applications - Codechef
- Analysis and Design of Algorithm - Codechef
- Problem Solving Through Programming in C - NPTEL
- Programming, Data Structures and Algorithms - NPTEL
- Data Analysis with Python - IBM
- Python for Data Science - IBM

SKILLS

- Language - PYTHON
- Data Analysis - Numpy | Pandas | Matplotlib | Seaborn | PowerBI | Tableau | Alteryx
- Machine Learning - Scikit-Learn | Tensorflow | Keras | PyTorch
- LLM Frameworks - Langchain | Langgraph | LangSmith | MCP Client | MCP Server
- Backend - FastAPI | Pydantic
- Database - MySQL | Microsoft SQL Server
- Vector DB - pgvector | FAISS
- Tools - Git | Docker | VS Code | Jupyter Notebook | Anaconda | Replit
- Cloud - AWS

LEADERSHIP & ACTIVITIES

- Hackathon Participation – Explored innovative problem-solving approaches and gained hands-on experience in collaborative development.
- Python Mentor, Pynocturnal Club – Conducted sessions to teach Python fundamentals and problem-solving to juniors.
- Event Lead, NVIDIA TechVerse – Led and organized the event, managing coordination and ensuring smooth execution.

PUBLICATIONS

- Shantveer Kesti, “A Survey on Fast Task Adaptation Techniques in Machine Learning”, IEEE Xplore, May 2025. Link
- Shantveer Kesti,“SleepSensei-AI: Revolutionizing Sleep Coaching”,IEEE Xplore, February 2026. Link